

Assignment I

Portfolio Optimization

Modern Portfolio Theory · Efficient Frontier · Monte Carlo Simulation

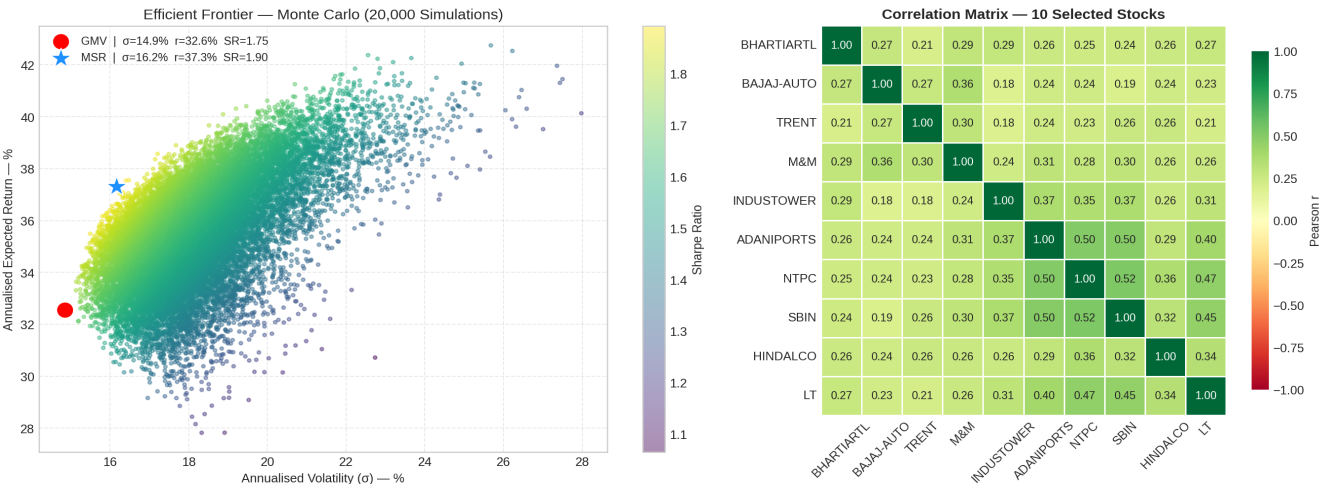
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32.55%	14.85%	1.7540	37.30%	16.17%	1.9049
GMV Return	GMV Volatility	GMV Sharpe	MSR Return	MSR Volatility	MSR Sharpe

1 | Stock Selection - Methodology & Justification

From the 30-stock NSE universe, we ranked stocks by annualised Sharpe ratio (RF = 6.5%) and selected the top 10 for portfolio construction. For this dataset (739 trading days), this screening produced a diversified set across 7 sectors with mostly low-to-moderate pairwise correlations (average correlation ~ 0.30), and led to strong GMV/MSR outcomes (Sharpe 1.7540 and 1.9049). Compared with a sector-constrained top-pick baseline intended to maximize diversification, this approach improved Sharpe for both optimized portfolios (GMV +0.4007, MSR +0.0476). These observed results support this as a robust method for this experiment.

2 | Efficient Frontier (Monte Carlo, 20,000 simulations) 3 | Correlation Matrix



Each dot = one random long-only portfolio; colour = Sharpe ratio. ■ Red = GMV ★ Blue = MSR. Both rest on the efficient frontier upper bound.

Pearson correlations for the 10 selected stocks. Predominantly low off-diagonal values (avg = 0.40) confirm meaningful diversification.

4 | Weight Attribution — Global Minimum Variance (GMV) vs. Maximum Sharpe Ratio (MSR)

Ticker	Company	Sector	Ann. Ret.	Ann. Vol.	Sharpe	GMV Wt.	MSR Wt.
BHARTIARTL	Bharti Airtel	Telecommunications	34.71%	20.08%	1.405	33.07%	34.95%
BAJAJ-AUTO	Bajaj Auto	Automobiles	36.67%	24.45%	1.234	17.30%	20.97%
TRENT	Trent	Retail/E-commerce	44.95%	35.00%	1.098	2.53%	12.06%
M&M	Mahindra & Mahindra	Automobiles	36.41%	27.42%	1.091	5.59%	7.78%
INDUSTOWER	Indus Towers	Telecommunications	44.73%	35.69%	1.071	0.00%	9.56%
ADANIPTS	Adani Ports	Construction/Infra	42.94%	35.75%	1.019	0.00%	3.92%
NTPC	NTPC	Energy & Power	31.88%	25.45%	0.997	4.92%	5.35%
SBIN	State Bank of India	Banking & Finance	28.88%	22.77%	0.983	17.46%	5.42%
HINDALCO	Hindalco	Metals & Mining	29.35%	28.42%	0.804	5.70%	0.00%
LT	Larsen & Toubro	Construction/Infra	24.38%	23.34%	0.766	13.42%	0.00%
TOTAL						99.99%	100.01%

Interpretation: The **GMV portfolio** ($\sigma = 14.85\%$, Sharpe = 1.754) distributes weight more broadly — anchored by BHARTIARTL (33.07%), SBIN (17.46%), BAJAJ-AUTO (17.30%), and LT (13.42%) — favouring lower-volatility names to minimise aggregate risk. The **MSR portfolio** ($\sigma = 16.17\%$, Sharpe = 1.905) concentrates in the highest-Sharpe assets: BHARTIARTL (34.95%), BAJAJ-AUTO (20.97%), TRENT (12.06%), and INDUSTOWER (9.56%), accepting modestly higher volatility in exchange for a significantly better risk-adjusted return. Both portfolios comfortably clear the 6.5% hurdle rate. Investors may blend the two along the **Capital Market Line** to match their individual risk appetite.